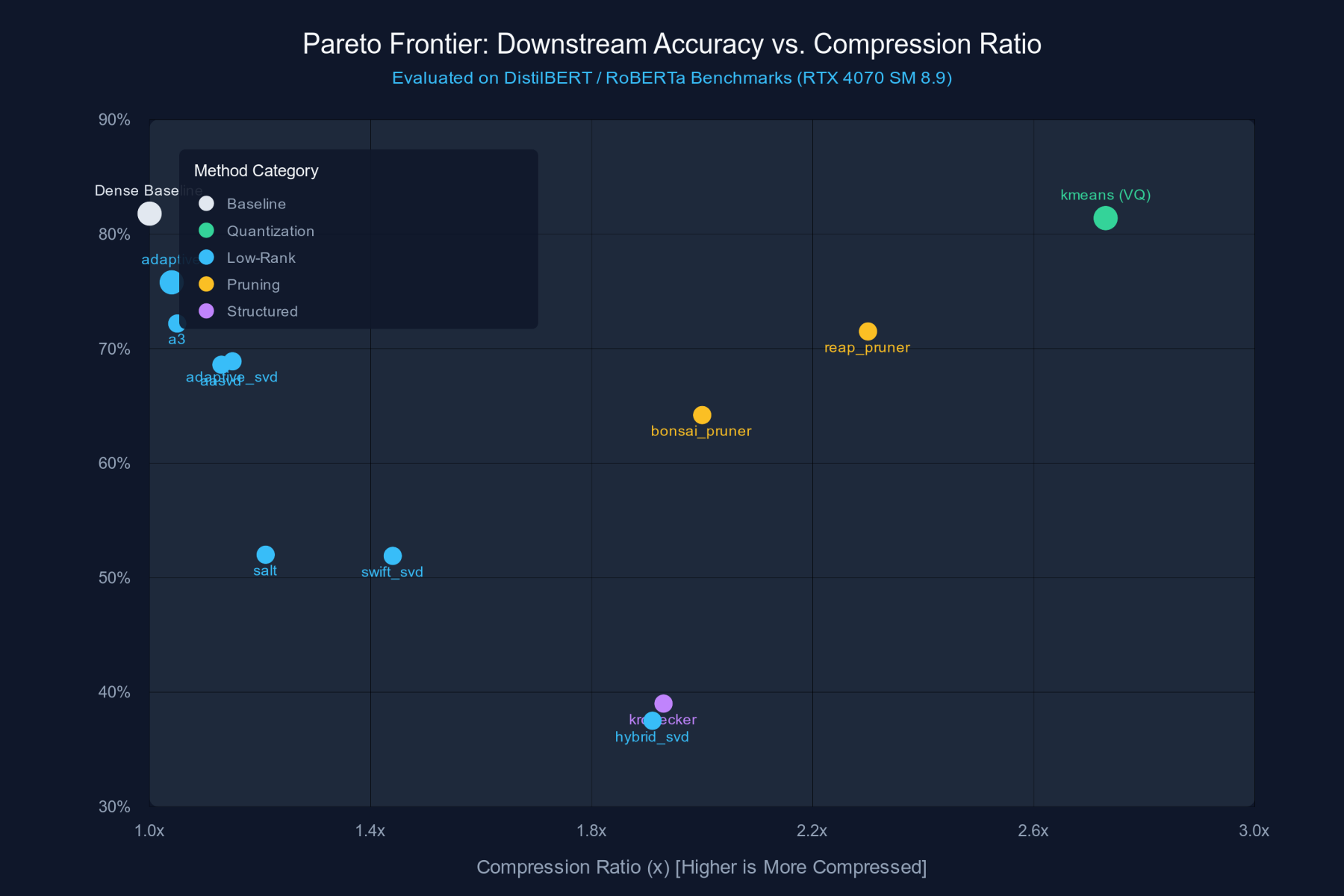

Pareto Frontier: Downstream Accuracy vs. Compression Ratio
Evaluated on DistilBERT / RoBERTa Benchmarks (RTX 4070 SM 8.9)
Method Category
Baseline
Quantization
Low-Rank
Pruning
Structured
Dense Baseline
kmeans (VQ)
adaptive
a3
reap_pruner
adaptive_svd
aasvd
bonsai_pruner
salt
swift_svd
kronecker
hybrid_svd
90%
80%
70%
60%
50%
40%
30%
1.0x
1.4x
1.8x
2.2x
2.6x
3.0x
Compression Ratio (x) [Higher is More Compressed]